Cloud Security and Trust Management

Agenda

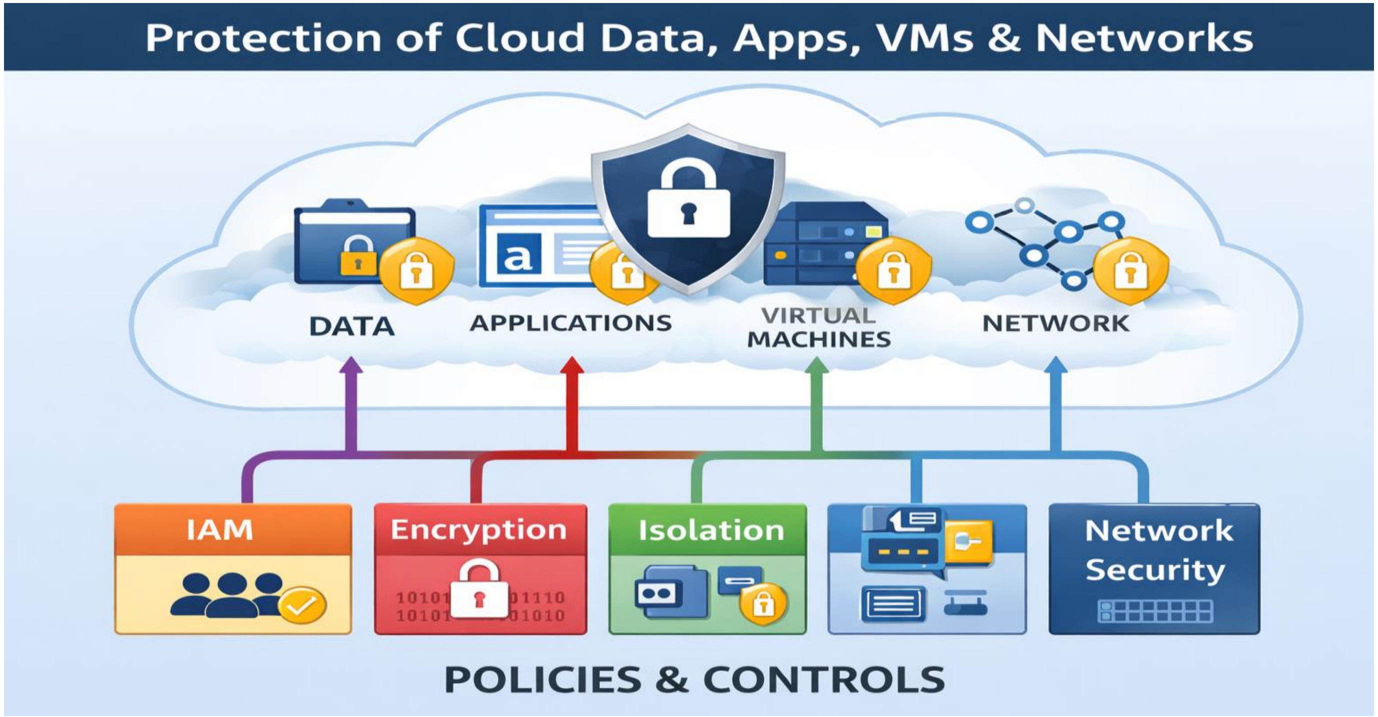
1. Introduction
2. Cloud Threats
3. Security Architecture
4. Core Security
5. Trust Management
6. Case Study
7. Best Practices

Learning Objectives

- Understand cloud security
- Identify threats
- Learn security mechanisms
- Understand trust models

What is Cloud Security?

- Protection of cloud data, apps, VMs and networks using policies & controls.



Why Cloud Security Matters

- Shared infrastructure
- Remote access
- Third party providers
- Large attack surface
  - Traditional Server Application: 1 Web app ,1 Database, 2 open ports
  - Cloud Application: (Web frontend, Backend API,Admin portal ,Kubernetes dashboard, SSH to VMs,Object storage ,CI/CD pipeline,Monitoring endpoints

Cloud Threat Landscape

- Data Breach
- Insecure APIs
- Account Hijacking
- Insider Threats
- DDoS
- Misconfiguration

Example: Data Breach

- Public storage exposure → sensitive data leak
- Impact: Legal + Reputation loss

Cloud Security Architecture

- User → IAM → Firewall → VM → Storage
- Security applied at every layer

Shared Responsibility Model

Cloud Provider responsibilities:

- Physical security of data centers
  - Controlled, need-based access
- Hardware and software infrastructure
  - Storage decommissioning,
  - host operating system (OS) access logging,
  - and auditing
- Network infrastructure
  - Intrusion detection
- Virtualization infrastructure
  - Instance isolation

Customer responsibilities:

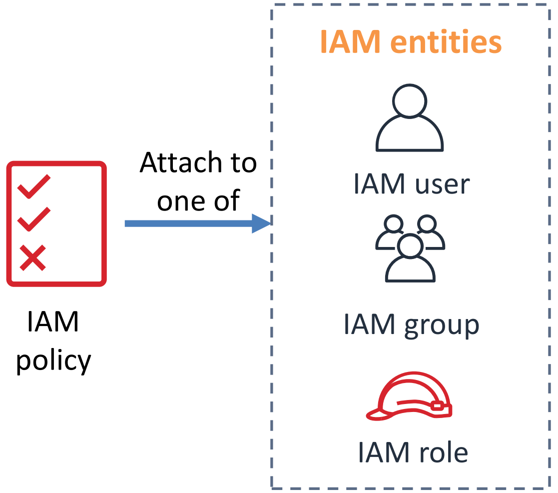
- Amazon Elastic Compute Cloud (Amazon EC2) instance operating system
  - Including patching, maintenance
- Applications
  - Passwords, role-based access, etc.
- Security group configuration
- OS or host-based firewalls
  - Including intrusion detection or prevention systems
- Network configurations
- Account management
  - Login and permission settings for each user

Identity & Access Management

- Authentication
- Authorization
- RBAC roles

IAM policies

- **An IAM policy is a document that defines permissions**
  - Enables fine-grained access control
- Two types of policies – *identity-based* and *resource-based*
- **Identity-based** policies –
  - Attach a policy to any IAM entity
    - An IAM user, an IAM group, or an IAM role
  - Policies specify:
    - Actions that ***may*** be performed by the entity
    - Actions that ***may not*** be performed by the entity
  - A single *policy* can be attached to multiple *entities*
  - A single *entity* can have multiple *policies* attached to it
- **Resource-based** policies
  - Attached to a resource (such as an S3 bucket)



User based

- {
- "Version": "2012-10-17",
- "Statement": [
- {
- "Effect": "Allow",
- "Action": [
- "ec2:StartInstances",
- "ec2:StopInstances"
-],
- "Resource": "*"
- }
-]
- }

Resource based (S3)

- {
- "Version": "2012-10-17",
- "Statement": [
- {
- "Effect": "Allow",
- "Principal": {
- "AWS": "arn:aws:iam::123456789012:user/Alice"
- },
- "Action": "s3:GetObject",
- "Resource": "arn:aws:s3:::student-data/*"
- }
-]
- }

Data Security

- Encryption at Rest & Transit
- Key Management Services

Virtualization Security

- VM isolation
- Secure Hypervisor
- Container Sandboxing

Network Security

- VPC
- Firewalls
- Security Groups
- IDS/IPS

What is Trust?

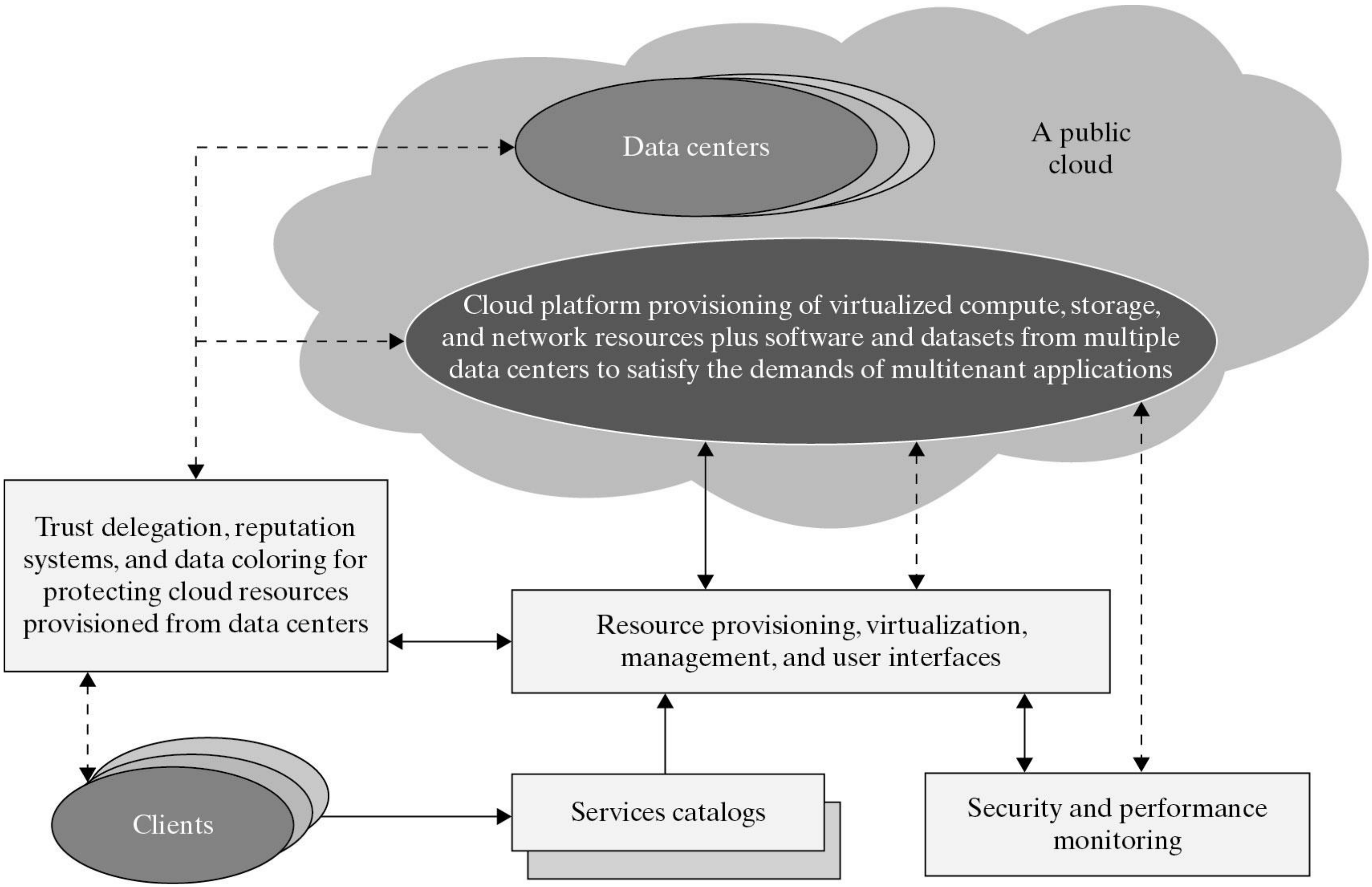
- Confidence in provider behavior, data safety and SLA compliance

$Ex' = \overline{X}$

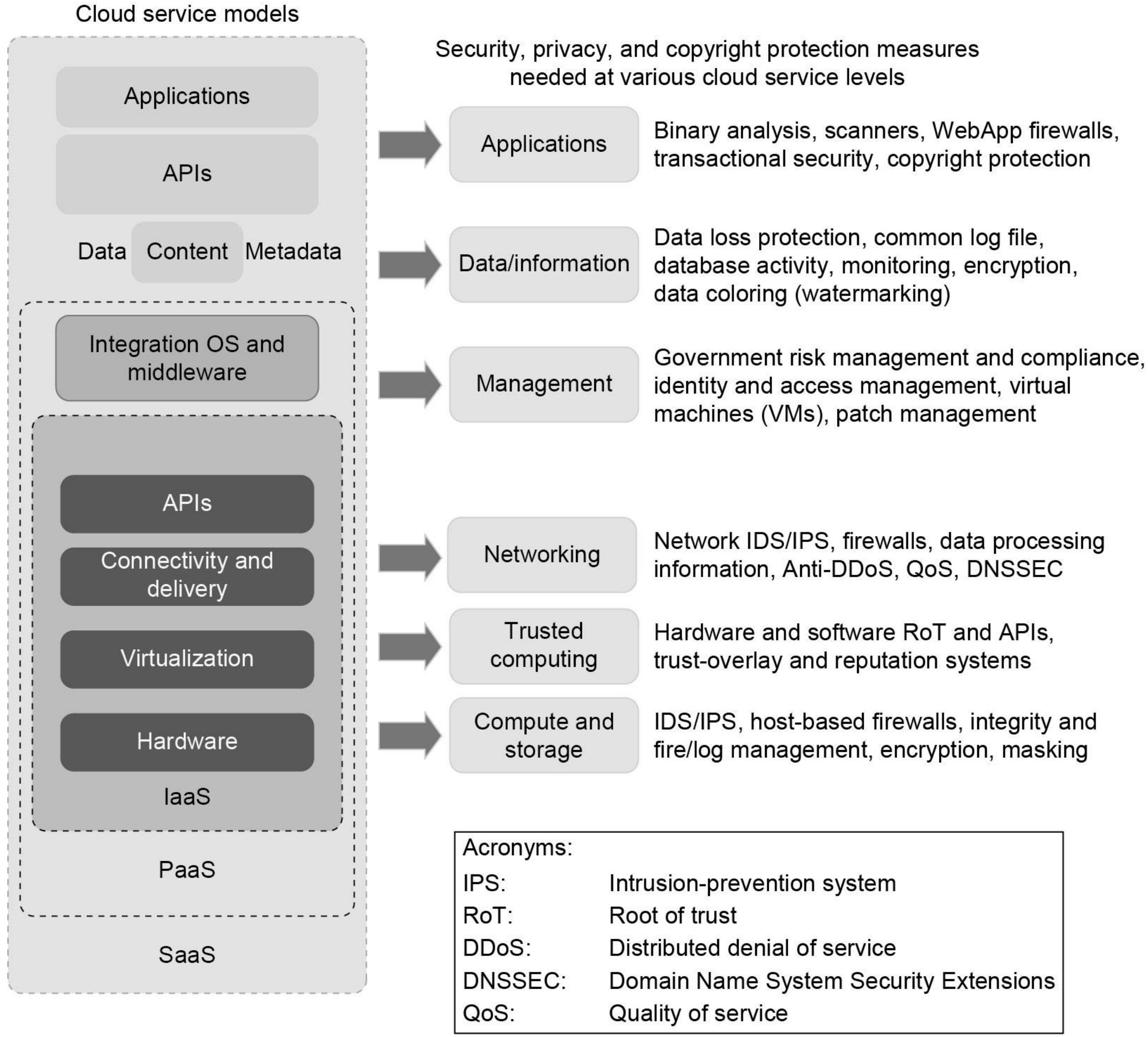
Security and Trust Barriers in Cloud Computing

- Protecting datacenters must first secure cloud resources and uphold user privacy and data integrity.
- Trust overlay networks could be applied to build reputation systems for establishing the trust among interactive datacenters.
- A watermarking technique is suggested to protect shared data objects and massively distributed software modules.
- These techniques safeguard user authentication and tighten the data access-control in public clouds.
- The new approach could be more cost-effective than using the traditional encryption and firewalls to secure the clouds.

Security Aware Cloud Platform



Cloud Service Models & Security Measures



(a) Cloud service models

(b) Security, privacy, and copyright protection measures

Trust Management Components

- Establish → Evaluate → Update → Revoke

Trust Models

- Reputation based
- Policy based
- Evidence based

Trust Example

- Select provider based on uptime + certification + ratings

Case Study: Capital One

- Firewall misconfiguration
- 100M records leaked

Best Practices

- Strong IAM
- Encryption
- Monitoring
- Zero Trust
- Audits

Zero Trust Model

- Never trust, always verify

Summary

- Security + Trust = Safe Cloud Adoption

Thank You

- Questions?